

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12384491
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Shipman; Christopher et al.

Bicycle derailleur

Abstract

A bicycle derailleur includes a base member mountable to a bicycle frame and a cage assembly moveably coupled to the base member. One or both of a motor and/or electrical generator system may be coupled to and moveable with the cage assembly. The motor is operable to move the cage assembly. The electrical generator system includes a generator and a generator drive system. An energy storage device, spaced apart from the cage assembly, may be connected to the electrical generator system and/or motor with a flexible electrical conductor.

Inventors:	Shipman; Christopher (Chicago, IL), Hahn; Sage (Chicago, IL)
Applicant:	SRAM, LLC (Chicago, IL)
Family ID:	83901940
Assignee:	SRAM, LLC (Chicago, IL)
Appl. No.:	17/867127
Filed:	July 18, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220355899 A1	Nov. 10, 2022

Related U.S. Application Data

continuation-in-part parent-doc US 17314949 20210507 US 11713095 child-doc US 17867127

Publication Classification

Int. Cl.: B62M9/124 (20100101); B62M9/122 (20100101); B62M9/126 (20100101); B62M9/128 (20100101)

U.S. Cl.:

CPC **B62M9/124** (20130101); **B62M9/122** (20130101); **B62M9/126** (20130101); **B62M9/128** (20130101); B62M2009/12406 (20130101)

Field of Classification Search

CPC: B62M (9/1248); B62M (9/1242); B62M (9/126); B62M (9/122); B62M (9/128); B62M (25/08); B62M (9/121); B62M (9/124); B62M (2009/12406); B62M (2009/12413); B62M (9/125); B62M (25/00); B62M (9/131); B62M (9/132); B62M (9/134); B62M (9/1342); B62M (9/135); B62M (9/136)

USPC: 474/80; 474/82

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5470277	12/1994	Romano	N/A	N/A
5480356	12/1995	Campagnolo	N/A	N/A
7042123	12/2005	Kitamura	280/260	H02K 7/1846
8721495	12/2013	Kitamura	477/7	B62M 25/08
10220913	12/2018	Kitamura	N/A	N/A
10793222	12/2019	Harris	N/A	B62M 9/1244
11608139	12/2022	Komatsu	N/A	B62M 9/122
11713095	12/2022	Shipman	474/80	B62M 9/124
2004/0232880	12/2003	Nyachoto	320/104	H02J 7/1407
2005/0043129	12/2004	Guderzo	474/70	B62M 25/08
2012/0322591	12/2011	Kitamura	474/80	B62M 9/122
2012/0322594	12/2011	Kitamura	474/110	B62M 9/122
2015/0111675	12/2014	Shipman	474/82	B62M 9/122
2017/0113759	12/2016	Watarai	N/A	B62J 43/30
2018/0229803	12/2017	Wesling	N/A	B62M 9/132
2018/0237104	12/2017	Pasqua	N/A	B62M 9/122
2018/0354586	12/2017	Komatsu	N/A	N/A
2019/0100279	12/2018	Brown	N/A	N/A
2019/0351971	12/2018	Dueweling	N/A	B62M 9/122
2020/0166089	12/2019	Choltco-Devlin et al.	N/A	N/A
2020/0269954	12/2019	Mizutani	N/A	N/A
2020/0339220	12/2019	Boehm	N/A	N/A
2020/0361565	12/2019	Komatsu	N/A	B62M 9/122
2022/0355900	12/2021	Shipman	N/A	B62M 9/1242
2023/0002006	12/2022	Kok	N/A	B62M 25/08
2023/0312054	12/2022	Shipman	474/80	B62M 9/1248

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
4109006	12/1991	DE	N/A
202015003858	12/2014	DE	B62M 9/122
102019204885	12/2019	DE	N/A

2535254	12/2011	EP	N/A
2535257	12/2011	EP	N/A
201325034	12/2012	TW	N/A
I644831	12/2017	TW	N/A
202108439	12/2020	TW	N/A
202108443	12/2020	TW	N/A
9526900	12/1994	WO	N/A
2015068125	12/2014	WO	N/A

Primary Examiner: Momper; Anna M

Assistant Examiner: Dias; Raveen J

Background/Summary

(1) This application is a continuation-in-part of U.S. application Ser. No. 17/314,949, filed May 7, 2021 and entitled “Bicycle Derailleur,” the entire disclosure of which is hereby incorporated herein by reference.

FIELD OF THE DISCLOSURE

(1) The present application generally relates to a bicycle derailleur, including for example and without limitation a bicycle rear derailleur incorporating an energy harvesting system.

BACKGROUND

(2) Bicycle derailleurs are commonly incorporated into a drivetrain of a bicycle. The typical drivetrain also includes a crank assembly that is coupled to one or more drive sprockets. The crank assembly is operable to drive a chain that is routed or wrapped around one of the drive sprockets. The chain may also be routed to one of wheels of the bicycle, for example a rear wheel, wherein the chain may engage one or more driven sprockets. Derailleurs are provided as a part of the drivetrain. For example, one derailleur (e.g., front) may be located adjacent the one or more drive sprockets, while another derailleur (e.g., rear) may be located adjacent the driven sprockets, for example adjacent the rear wheel. The derailleur(s) may be actuated to selectively shift the chain of the drivetrain between the drive sprockets, and/or to selectively shift the chain between the one or more of the driven sprockets. Shifting of the bicycle chain from one drive sprocket to another, or from one driven sprocket to another, is done in order to change the gear ratio of the drivetrain. The rear derailleur may also apply a tension to the chain to take-up slack, as well as to maintain the desired tension, in the chain on the non-drive-side of the drivetrain.

(3) In some embodiments, the rear derailleur may be a wireless, electrically actuated rear derailleur, which relies on a battery to supply power to a motor. The battery may need to be periodically charged or replaced, which may be an inconvenience if the battery is discharged during use of the bicycle, or at a remote location not conveniently located to a charging station or providing access to replacement batteries.

SUMMARY

(4) In one aspect, one embodiment of a bicycle derailleur may include a base member mountable to a bicycle frame and a cage assembly moveably coupled to the base member. The cage may be moveable in opposite first and second directions relative to the base member. A motor may be coupled to and moveable with the cage assembly in the opposite first and second directions. The motor may be operable to move the cage assembly in the opposite first and second directions. In one embodiment, the cage assembly may include a chain pulley rotatably coupled to the cage assembly about a first rotation axis.

(5) In another aspect, one embodiment of a bicycle derailleur may include a base member mountable to a bicycle frame and a cage assembly moveably coupled to the base member. An electrical generator system may be coupled to and moveable with the cage assembly. The electrical generator system may include a generator and a generator drive system. The generator drive system may include a chain pulley rotatably coupled to the cage assembly about a first rotation axis and a generator transmission operably coupled between the chain pulley and the generator.

(6) In another aspect, one embodiment of a bicycle derailleur may include a base member mountable to a bicycle frame and a cage assembly moveably coupled to the base member. An electrical generator system may be coupled to and moveable with the cage assembly. The electrical generator system may include a generator and a generator drive system. The generator drive system may include a chain pulley rotatable about a first rotation axis in opposite first and second rotational directions and a clutch. The clutch may driveably connect the chain pulley and the generator when the chain pulley is rotated in the first rotational direction such that the generator is activated. The clutch disconnects the chain pulley and the generator when the chain pulley is rotated in the second rotational direction such that the generator is deactivated.

(7) In another aspect, a bicycle derailleur may include a base member mountable to a bicycle frame and a cage assembly moveably coupled to the base member. An electrical generator system may be coupled to and moveable with the cage assembly. The electrical generator system may include a generator and a generator drive system. The generator drive system may include a chain pulley rotatable about a first rotation axis, at least first and second pulleys rotatable about second and third rotation axes respectively, and a belt engaged with the first and second pulleys. In one embodiment, the first and second pulleys may have a pulley ratio greater than 1.

(8) In another aspect, one embodiment of a bicycle derailleur includes a base member mountable to a bicycle frame, a cage assembly moveably coupled to the base member with a coupling assembly, and an electrical generator system coupled to and moveable with the cage assembly. In one embodiment, at least one energy storage device is mounted on at least one of the base member and/or coupling assembly, wherein the energy storage device is electrically connected to the electrical generator system. In another embodiment, an energy storage device is spaced apart from the cage assembly, wherein the cage assembly is moveable relative to the energy storage device. A flexible electrical conductor electrically connects the energy storage device and the electrical generator system.

(9) The various aspects and embodiments of the derailleur, and the methods for the use and assembly thereof, may provide significant advantages over other derailleurs and methods. For example and without limitation, the motor and/or the electrical generator system may be mounted on and moveable with the cage. If damaged, the cage, including the motor and electrical generator system, may be quickly and easily replaced without having to replace the other components of the derailleur. In addition, the cage, which may include a chain pulley engageable by a chain, provides an input to the generator system and motor, with the components being positionally fixed relative to each other and moveable with the cage as the cage is: (1) moved laterally during a gear changing sequence, and/or (2) rotated to maintain a tension in the chain. In this way, the assembly avoids the need for couplings, whether electrical or mechanical, between any components located on the cage and components located on other parts of the derailleur, which may be moveable relative to each other. In addition, the electrical generator system ensures that power is always available to power the motor, for example during shifting, and/or for other activities and accessories requiring electrical power. In addition, an energy storage device may be spaced apart from the cage and be electrically connected to the electrical generator system, for example with an electrical conductor, such that additional energy storage options, including batteries, rechargeable and non-rechargeable, may be available to power the motor and/or other accessories.

(10) The foregoing paragraphs have been provided by way of general introduction and are not intended to limit the scope of the claims presented below. The various preferred embodiments,

together with further advantages, will be best understood by reference to the following detailed description taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Objects, features, and advantages of the present invention will become apparent upon reading the following description in conjunction with the drawing figures, in which:
- (2) FIG. 1 is a partial side view of a bicycle assembled with a rear derailleur having a cage assembly in a fully rotated clockwise position.
- (3) FIG. 2 is a partial side view of a bicycle assembled with a rear derailleur having a cage assembly in a fully rotated counterclockwise position.
- (4) FIG. 3 is a top view of a rear derailleur in an extreme outboard position.
- (5) FIG. 4 is an outboard side view of one embodiment of a rear derailleur having a cage assembly in a fully rotated clockwise position.
- (6) FIG. 5 is a top view of a rear derailleur in an extreme inboard position.
- (7) FIG. 6 is an outboard side view of one embodiment of a rear derailleur having a cage assembly in a fully rotated counterclockwise position.
- (8) FIG. 7 is a side perspective view of a cage assembly.
- (9) FIG. 8 is an outboard side view of the cage assembly shown in FIG. 7.
- (10) FIG. 9 is an inboard side view of the cage assembly shown in FIG. 7.
- (11) FIG. 10 is a cross-sectional view of the cage assembly taken along line 10-10 in FIG. 8.
- (12) FIG. 11 is a perspective view of a chain pulley.
- (13) FIG. 12 is a perspective view of a chain pulley shaft.
- (14) FIG. 13 is a cross-sectional view of the cage assembly taken along line 13-13 in FIG. 8.
- (15) FIG. 14 is a perspective view of a belt tension adjustment system.
- (16) FIG. 15 is a cross-sectional view of the cage assembly taken along line 15-15 in FIG. 8.
- (17) FIG. 16 is a perspective view of a main printed circuit board.
- (18) FIG. 17 is a perspective view of a motor and shifting drive system.
- (19) FIG. 18 is an exploded view of the motor and shifting drive system shown in FIG. 17.
- (20) FIG. 19 is a partial side perspective view of a cage assembly with a cover removed.
- (21) FIG. 20 is a cross-sectional view of the derailleur taken along line 20-20 in FIG. 4.
- (22) FIG. 21 is a cross-sectional view of the derailleur taken along line 21-21 in FIG. 6.
- (23) FIG. 22 is a perspective view of a cage shaft.
- (24) FIG. 23 is a perspective view of a lead nut.
- (25) FIG. 24 is a perspective view of a lead screw and spur gear.
- (26) FIG. 25 is a perspective view of a crank arm.
- (27) FIG. 26 is a perspective view of a drive element.
- (28) FIG. 27 is a perspective view of a pin.
- (29) FIG. 28 is a perspective view of a derailleur with a portion of a moveable member cut-away.
- (30) FIG. 29 is a cross-sectional view taken along line 29-29 in FIG. 5.
- (31) FIG. 30 is a partial side view of another cage assembly embodiment.
- (32) FIG. 31 is an opposite partial side view of the cage assembly shown in 30.
- (33) FIG. 32 is a perspective view of one embodiment of a gear configured with a clutch.
- (34) FIG. 33 is a perspective view of a drive member shown in FIG. 32 including a portion of the clutch.
- (35) FIG. 34 is a schematic illustrating the generator and motor control system.
- (36) FIGS. 35A and B are graphs showing respectively forward and reverse pedaling generator phase detection sequences.

- (37) FIG. 36 is a graph showing Sensed Voltage v. Time for a sequence of pedaling.
- (38) FIG. 37 shows a side view of a bicycle having a rear derailleur.
- (39) FIG. 38 is an outboard side view of an alternative embodiment of a rear derailleur having a cage assembly in a fully rotated clockwise position.
- (40) FIG. 39 is an outboard side view of the rear derailleur shown in FIG. 38 with the cover removed.
- (41) FIG. 40 is a perspective view of the rear derailleur shown in FIG. 38.
- (42) FIG. 41 is perspective view of the rear derailleur shown in FIG. 39.
- (43) FIG. 42 is a schematic illustration showing the various mounting options of the energy storage device relative to the rear derailleur.

DETAILED DESCRIPTION OF THE DISCLOSURE

(44) It should be understood that the term “plurality,” as used herein, means two or more. The term “longitudinal,” as used herein means of or relating to a length or lengthwise direction. The term “lateral,” as used herein, means situated on, directed toward or running in a side-to-side direction. The term “coupled” means connected to or engaged with, whether directly or indirectly, for example with an intervening member, and does not require the engagement to be fixed or permanent, although it may be fixed or permanent. The terms “first,” “second,” and so on, as used herein are not meant to be assigned to a particular component so designated, but rather are simply referring to such components in the numerical order as addressed, meaning that a component designated as “first” may later be a “second” such component, depending on the order in which it is referred. It should also be understood that designation of “first” and “second” does not necessarily mean that the two components or values so designated are different, meaning for example a first direction may be the same as a second direction, with each simply being applicable to different components. The terms “upper,” “lower,” “rear,” “front,” “fore,” “aft,” “vertical,” “horizontal,” “right,” “left,” “inboard,” “outboard” and variations or derivatives thereof, refer to the orientations of an exemplary bicycle 150, shown in FIG. 37, from the perspective of a user seated thereon, for example with an “inboard” component or feature being closer to a vertical mid-plane of the bicycle extending in a direction 201. The term “transverse” means non-parallel. The terms “outer” and “outwardly” refers to a direction or feature facing away from a centralized location, for example the phrases “radially outwardly,” “radial direction” and/or derivatives thereof, refer to a feature diverging away from a centralized location, for example a rotation axis 152 of the cassette 3 as shown in FIG. 2. Conversely, the terms “inward” and “inwardly” refers to a direction facing toward the centralized or interior location. The term “subassembly” refers to an assembly of a plurality of components, with subassemblies capable of being further assembled into other subassemblies and/or a final assembly, such as the bicycle 150.

(45) Bicycle:

(46) FIG. 37 illustrates one example of a human powered vehicle. In this example, the vehicle is one possible type of bicycle 150, such as a road bicycle. The bicycle 150 has a frame 2, handlebars 154 near a front end of the frame, and a seat or saddle 156 for supporting a rider over a top of the frame. The bicycle 150 has a first or front wheel 158 carried by a front fork subassembly 160 supporting the front end of the frame. The bicycle 150 also has a second or rear wheel 162 supporting a rear end of the frame 2, which includes a pair of chain stays 164 connected to a pair of seat stays 165 (see also FIG. 1). The rear end of the frame 2 may be supported by a rear suspension component, such as a rear shock 167. The bicycle 150 also has a drive train 168 with a crank assembly 166 that is operatively coupled via a bicycle chain 4 to a rear cassette 3, otherwise referred to as a driven sprocket assembly, near the hub providing a rotation axis of the rear wheel 162. The crank assembly 166 includes at least one, and typically two, crank arms 170 and pedals 176, along with a front chainring assembly 172, or drive sprocket assembly. A crank spindle or shaft may connect the two crank arms. The crank shaft defines a center rotational axis 174 of the chainring assembly 172. The crank assembly may also include other components.

(47) A rear gear change device, such as a rear derailleur **180**, is disposed at the rear wheel **162** to move the bicycle chain **4** to different sprockets of the cassette **3**. In one embodiment, a front gear changer device, or front derailleur, may be provided to move the chain **4** to different sprockets of the chainring assembly. In the illustrated example, the saddle **156** is supported on a seat post **178** having an end portion received in a top of a frame seat tube of the frame **2**.

(48) In FIG. **37**, a normal riding or forward moving direction **201** of the bicycle **150** is shown. While the bicycle **150** depicted in FIG. **37** is a mountain bicycle, the rear gear change device, or rear derailleur **180**, including the specific embodiments and examples disclosed herein as well as alternative embodiments and examples, may be implemented on other types of bicycles. For example and without limitation, the disclosed rear derailleur **180** may be used on road bicycles, and any other type of bicycle incorporating a derailleur. It should be understood that the various energy harvesting systems and embodiments disclosed herein may also be incorporated into derailleurs, at any location (e.g., front), having a cage assembly with a rotatable wheel.

(49) Rear Derailleur:

(50) Referring to FIGS. **1-6**, **28** and **38-41**, the rear derailleur **180** includes a base member **5**, otherwise referred to as b-knuckle, which may be attached to the bicycle frame **2** with a fastener **10**, for example a b-bolt. The frame **2** may include a hanger **184** connected to the frame **2** at a junction between the seat stay **165** and the chain stay **164**. The base member **5** may be connected to the hanger frame **184**, or directly to the frame **2**. The base member **5** is removably coupled to the frame **2** with the fastener **10**, and may be free to pivot around an axis (A) of the fastener **10**. An adjustment member **11**, which may be configured as a screw, is threadably engaged with the base member, and may be actuated so as to adjust the rotational position of the base member **5** with respect to frame **2**.

(51) A linkage **186** includes an inner link **7** and an outer link **6** having first ends **188**, **190** that are rotatably connected to the base member **5**, for example with a first axle or pin **15** and a second axle or pin **16**, respectively. A moveable member **8**, otherwise referred to as a p-knuckle, is rotatably connected to opposite second ends **194**, **196** of the inner link **7** and the outer link **6** with a third axle or pin **17** and a fourth axle or pin **18**, respectively, such that the links **6**, **7** extend between the base member **5** and moveable member **8**. The pin **18** may be supported by a pair of bushings **103** in the moveable member **8** as shown in FIG. **29**. The base member **5**, moveable member **8**, inner link **7**, and outer link **6** form a four-bar linkage, and in particular a parallelogram linkage **186**. In this way, the moveable member **8** may be moveably coupled to the base member **5**. In one embodiment, the moveable member **8** is moveable relative to the base member **5**, and the frame **2**, in opposite inboard and outboard directions. It should be understood that the moveable member **8** may be moveably connected to the base member **5** with other components and/or linkages other than the disclosed linkage, which may or may not include one or more links.

(52) A biasing member **14**, which may be configured as a torsion spring, biases the linkage **186** to rotate clockwise, or outboard, as shown in FIG. **3**, for example about the first and second pins **15**, **16**. In one embodiment, the biasing member **14** may be arranged co-axially with second pin **16** and have a first end engaging the base member and a second end engaging the linkage **186**, for example the outer link **6**. The derailleur may include a rotational limit, including for example an upper limit screw **12** and a lower limit screw **13**, which are threadably engaged with base member **5** and limit the movements of the inner link **7** and the outer link **6**, respectively.

(53) In an alternative embodiment, the biasing force of the biasing member **14** may be reversed, such that the biasing member **14** bias the linkage **186** to rotate counterclockwise, or inboard, with reference to FIG. **3**. In operation, the chain **4** may apply a drag force to a pulley wheel **22**, which may cause the linkage **186** to want to rotate counterclockwise as shown in FIG. **3**. In this embodiment, the biasing member **14** and pulley wheel **22** work together in tandem to bias the linkage **186** in the same direction. As such, the parallelogram linkage is biased in the same rotational direction by the biasing force applied by the biasing member **14** and the drag force

applied by the chain to the pulley wheel **22**.

(54) Referring to FIGS. **7-9** and **38-41**, the derailleur **180** may include a cage assembly **9** moveably coupled to the base member **5** with a coupling assembly **500**, which includes the linkage **186** and the moveable member **8** in one embodiment. The cage assembly **9** is moveable in opposite first and second translation directions (e.g., inboard and outboard) relative to the base member **5**. The cage assembly **9** may also be moveable in opposite first and second rotational directions (e.g., clockwise and counterclockwise rotation) about a laterally extending axis **B** relative to the moveable member **8** and the base member **5**, or moveable relative to the base member **5** with a combination of translation and rotation. In particular, the cage assembly **9** is rotatably connected to the moveable member with a fastener extending in a lateral direction and defining the rotation axis **B**. The cage assembly may rotate clockwise around the axis **B** of the fastener to take up slack in the chain **4**, which is engaged with the cassette **3**, an upper chain pulley **21** and a lower chain pulley **22**. The upper and lower chain pulleys **21**, **22** are rotatably connected to the cage assembly **9**.

(55) Referring to FIGS. **5-9**, **10**, **13** and **38-41**, the cage assembly **9** may include one or both of an outer cage **24** secured to an inner cage **25**, for example with fasteners **29**, **30**, configured as screws, a snap-fit, and/or other known suitable securing devices. The lower chain pulley **22** rotates on a ball bearing having an inner race clamped between the inner cage **25** and the outer cage **24** with the fasteners, configured as a screw **30**, threaded into the inner cage **25**. This connection also serves to fix the inner cage **25** to the outer cage **24**. A cover **23** is fixed to the outer cage **24** with a plurality of fasteners **28**, shown as nine screws, spaced apart around the perimeter of the cover **23**, with the cover and outer cage defining a housing **200**. It should be understood that the term “housing” refers to a component capable of supporting or holding other components, and may be enclosed or open. As such, the outer cage and/or cover may each individually define a housing, or may in combination define the housing **200**, which is enclosed and sealed to prevent the intrusion of fluid into an interior of the housing in one embodiment. Referring to FIGS. **10** and **13**, an elastomeric sealing element **32** may be disposed in a groove **202** formed in the outer cage **24**. It should be understood that the groove may alternatively be formed in the cover. The groove **202** may define a closed loop that follows the contour of the perimeter of the cover **23**, with the sealing element **32** forming a fluid-tight seal between the outer cage **24** and cover **23**, or housing **200**. In one embodiment, the fastener **29** is threadably engaged with outer cage **24**, and fixes the inner cage **25** to the outer cage **24**.

(56) As shown in FIGS. **10-13**, one embodiment of the upper wheel pulley assembly include a sealed ball bearing assembly **33** received in a recess **204** formed in the outer cage **24** and engaged by the shoulder of an annular flange **206**. A ball bearing assembly **34** is received in a recess **208** formed in the cover **23**, and is supported on an outer side surface **210** of the housing. A pulley shaft **35** is disposed inside, and supported by, the inner races of ball bearing assemblies **33** and **34**. The pulley shaft **35** has an end **212** rotatably supported by the ball bearing assembly **34**, with an annular shoulder **214** engaging the inner race and trapping the bearing assembly **34** between the shoulder **214** and the surface **210**. The pulley shaft **35** includes an annular flange **216** spaced laterally from the annular shoulder **214**, with the annular flange **216** defining a second annular shoulder **218** engaging the inner race of the ball bearing **34** and trapping the bearing assembly **33** between the shoulder **218** and the annular flange **206**. The annular shoulders **214**, **216** face in opposite directions such that the annular shoulders limit the axial movement of the shaft **35** in both lateral directions (inboard and outboard). The ball bearing assembly **33** is captured between the outer cage **24**, or an annular shoulder defined thereby, and the annular shoulder **216** of the shaft **35**. A sealing element **346**, configured in one embodiment as an O-ring, is located in a circumferential groove **220** formed in the shaft **35** inboard and spaced apart from the annular flange **216**. The groove may alternatively be formed in the ball bearing assembly. The sealing element **346** forms a fluid-tight seal between the pulley shaft **35** and the inner race of the ball bearing assembly **33**, such that fluid is prevented from entering the interior space of the housing **200** defined between the cover **23** and

the outer cage **24**. The pulley shaft **35** may include an axially extending hole **222**. The hole **222** may be threaded and threadably engaged by a fastener **31**, shown as a screw.

(57) Referring to FIGS. **11** and **12**, the upper chain pulley **21** may include a key hole **21b**, having a cross-shaped recess **226** in one embodiment. The pulley shaft **35** may be configured with a corresponding key **224**, or insert portion configured with mating corresponding cross-shaped protrusions **35b**. When the upper chain pulley **21** is assembled as shown in FIG. **10**, the key **224** interfaces and ensures a non-rotatable engagement or coupling with the key hole **21b**, for example by way of the engagement of the recesses **226** by the protrusions **35b**. The chain pulley **21** includes a hub **228** having an inner surface **230** that abuts an end surface **35a** of the pulley shaft **35** defined between the protrusions **35b**. By way of the interface between the key **224** and key hole **21b**, the upper chain pulley **21** is non-rotatably and axially secured to the pulley shaft **35**, such that torsion may be transmitted between the upper chain pulley **21** and the pulley shaft **35** via the key **224** and key hole **21b**, i.e., between the protrusions **35b** and recesses **226**. As shown in FIG. **10**, the pulley shaft **35** is captured between the cover **23** and the outer cage **24**, with the chain pulley **21** secured to the shaft **35**. As such, the chain pulley **21** is not captured or clamped between the inner and outer cages **25**, **24**. Rather, the chain pulley **21** is supported in a cantilevered fashion on an end of the shaft **35**, which is supported by the housing **200** outboard of the chain pulley **21**. As such, the inner cage **25** may not overlie the chain pulley **21**, and in particular the pulley shaft **35**, but rather may be configured to provide clearance between the inner cage **25** and the upper pulley system as shown in FIG. **9**. To replace the chain pulley, the inner cage **25**, therefore, does not need to be removed from the outer cage **24**. Of course, in other embodiments, the inner cage may overlie the chain pulley.

(58) Electrical Generator System:

(59) Referring to FIGS. **13-16**, **19**, **30**, **31** and **38-41**, an electrical generator system **240** is coupled to and moveable with the cage assembly **9**, meaning the electrical generator system **240** translates and rotates with the cage assembly **9** as the cage assembly moves relative to the moving member **8**, base **5** and bicycle frame **2**. Put another way, the electrical generator system is fixed to the cage assembly and follows the movement path of the cage assembly. In one embodiment, the electrical generator system **240** is disposed in the housing **200**, defined by the cover **23** and outer cage **24**, and may be entirely enclosed within a sealed interior cavity **242** defined between the cover **23** and outer cage **24**. The electrical generator system **240** includes a generator **50** and a generator drive system **244**, with the generator drive system **244** including an input. The input may be configured in one embodiment as the chain pulley **21** rotatably coupled to the housing **200**, and the outer cage **24** and cover **23** in particular, as shown in FIGS. **4-9** and **13**. The chain pulley **21** is rotatable about a rotation axis **246**. In an alternative embodiment, shown in FIGS. **38-41**, the input may be configured as the chain pulley **22** rotatably coupled to the housing. In this way, the generator **50** may be mounted to the cage between the upper and lower chain pulleys **21**, **22**, such that the generator may be driven by either (or both) of the chain pulleys **21**, **22**, or inputs. The generator drive system **244** may further include a generator transmission **250** operably coupled between the input, e.g., chain pulley **21** or chain pulley **22**, and an output, e.g., the generator **50**. The generator transmission **250** includes a first generator spur gear **37**, which is non-rotatably and axially fixed to the pulley shaft **35**, such that the spur gear **37** rotates with the chain pulley **21** about the rotation axis **246**, or with the chain pulley **22** about a rotation axis **247**. For example and without limitation, the spur gear **37** may be fixed to the pulley shaft with a friction fit, and/or with a spline interface. The spur gear **37** is disposed in a space, or interior cavity **242**, defined between the cover and outer cage. The first generator spur gear **37** has a plurality of teeth **260** spaced circumferentially around the perimeter of the spur gear, with the teeth **260** engaged with teeth **262** spaced circumferentially around the perimeter of a pinion gear **38**. The pinion gear **38** is rotatable about an axle **44** supported by and between the outer cage **24** and cover **23**. In one embodiment, the spur gear **37** has thirty (30) teeth **260**, while the pinion gear **38** has ten (10) teeth **262**, providing a gear ratio of 3:1, meaning that one rotation of the spur gear **37** results in three (3) rotations of the pinion gear **38**. In one

embodiment, the chain pulley **21** has a plurality of teeth, for example **12**. In one embodiment, the axle **44** has a first end engaged with the outer cage **24**, for example a non-rotatable engagement, e.g., threaded, such that the axle **44** is non-rotatable, with an opposite second end engaged with the cover **23**. Alternatively, the axle **44** may be allowed to rotate relative to the outer cage **24**. The pinion gear **38** includes a hub **264** that rotates on the axle **44** about a rotation axis **266**. A second generator spur gear **39** is co-axially and non-rotatably fixed to first generator pinion gear **38**, **338**, and to the hub **264** in particular, and rotates with and is driven by the pinion gear **38**, **338**. The spur gear **39**, **339** has thirty-six (36) teeth **268** spaced circumferentially around a perimeter thereof in one embodiment. The teeth **268** of the spur gear engage teeth **270** spaced circumferentially around the perimeter of a pinion gear **40**, which is supported by and rotatable about an axle **45** rotatably or non-rotatably supported by and between the outer cage **24** and cover **23**. In one embodiment, the axle **45** has a first end engaged with the outer cage **24**, for example a threadable engagement such that the axle is non-rotatable, with an opposite second end engaged with the housing **24**. In another embodiment, the axle is rotatably supported by and between the outer cage **24** and cover **23**. The pinion gear **40** includes a hub **274** that rotates on the axle about a rotation axis **272**. The pinion gear may have for example twelve (12) teeth **270**, providing a gear ratio of 3:1 between the spur gear **39** and pinion gear **40**, meaning that one rotation of the spur gear **39** results in three (3) rotations of the pinion gear **40**.

(60) The generator transmission **250** also includes at least first and second pulleys **41**, **42** rotatably coupled to the housing **200** about rotation axes **272**, **280** respectively, and a belt **43** engaged with the first and second pulleys. In one embodiment, a first generator pulley **41** preferably has a plurality of teeth **276**, for example 75 teeth **276**, and is co-axially and non-rotatably fixed to the hub **274** of the pinion gear **40**, with the pulley **41** and pinion gear **40** being rotatable about the rotation axis **272**. The second pulley **42** has a plurality of teeth **278**, for example 20 teeth, and is co-axially and non-rotatably fixed to a rotor **47** of the generator **50**, with the pulley **42** and rotor **47** being rotatable about the rotation axis **280**. A belt **43** forms a continuous or endless loop. In one embodiment, the belt **43** may be a toothed belt having a plurality of teeth **282** formed around an inner surface of the loop. In one embodiment, the belt **43** may include for example and without limitation 91 teeth **282**. The belt **43**, and the teeth **282** in particular, are engaged with the first and second pulleys **41**, **42** and the teeth **276**, **278** thereof. In one embodiment, the belt **43** is used at the location in the transmission where the rotational speed is the greatest, or the stage that directly rotates the generator. The belt may be used in place of gears, for example, to reduce the noise of the transmission because belts in general are quieter and don't create the high pitch sounds that very fast moving gear teeth generate. Of course, it should be understood that gears may be used in place of the belt. The pulleys **41**, **42** have a pulley ratio of greater than 1, and in one embodiment a ratio of 3.75:1. It should be understood that in other embodiments, the belt **43** and pulleys **41**, **42** may be configured without teeth, with the belt **43** engaging the pulleys **41**, **42** by way of friction. The rotor **47** is rotatably received in a bore **284** defined by a generator base **46**. The generator **50** includes a stator **48** disposed inside an outer periphery, or annular wall **286**, of the rotor **47**. The stator may be non-rotatably fixed to the generator base **46**, with the rotor **47** being rotatable relative to the stator **48** about the rotation axis **280**. In one embodiment, the chain pulley **21**, or input, has a first rotational speed and the generator **50**, e.g., the rotor **47** or output, has a second rotational speed, wherein the generator transmission provides a ratio between the first and second rotation speeds of between and including 20:1 and 50:1. In one embodiment, the overall gear ratio of the transmission between the upper pulley **21** and the rotor **47** of the generator **50** is preferably 33.75:1 (e.g., overall gear ratio=(gear ratio 1)*(gear ratio 2)*(pulley ratio 3)=3*3*3.75=33.75:1, meaning the rotational speed of rotor **47** will be 33.75 times greater than the rotational speed of upper pulley **21**. It should be understood that this ratio may be varied by changing any of the gear ratios between the spur gear **37** and the pinion gear **38**, **338**, the spur gear **39**, **339** and the pinion gear **40**, and the first pulley **41** and the second pulley **42**.

(61) In one embodiment, shown in FIGS. 30-33, the generator transmission 250 includes a clutch 290, which driveably connects the chain pulley 21 and the generator 50, and the rotor 47 in particular, when the chain pulley 21 is rotated in the first rotational direction such that the generator 50 is activated, meaning the rotor 47 is being rotated relative to the stator 48. The clutch 290 disconnects the chain pulley 21 and the generator 50 when the chain pulley 21 is rotated in the second rotational direction such that the generator 50 is deactivated, meaning the rotor 47 is not being rotated relative to the stator 48. In one embodiment, the clutch 290 is disposed between the spur gear 339 and the pinion gear 338. It should be understood that the clutch 290 may also be disposed between any coaxially mounted components, including for example between the chain pulley 21 and the spur gear 37, or the pinion gear 40 and the pulley 41. In other embodiments, the clutch may be disposed between non-coaxially mounted components, for example a slip interface between the adjacent pulleys. In one embodiment of the clutch, the pinion gear 338 defines a drive member, which is rotatably supported by the axle 44. It should be understood that the drive member may be a spur gear. In either case, the clutch 290 is disposed or mounted between the drive member, i.e., pinion gear 338, and a driven member, i.e., the spur gear 339. In this way, the clutch 290 provides for one-way rotational engagement between the drive member and the driven member. In one embodiment, the clutch 290 includes at least one spring finger 292, and preferably a plurality of spring fingers (shown as two). The driven member, or gear or pulley, may include in one embodiment a hub 294 having at least one indentation 296, and preferably a number of indentations 296 (e.g., 2) corresponding to the number of spring fingers 292, although the numbers may be different. The spring fingers 292, which are in-molded with the spur gear 339 in one embodiment, may be configured as cantilever leaf springs having a curved arm 302 extending from web of the spur gear 339 and terminating at an engagement end 304. The engagement end 304 is positioned radially inwardly from an inner bearing surface 306 of the driven gear, or spur gear 339, when the arm 302 is in a non-biased position, i.e., is not pre-loaded. The arm 302 is preloaded by moving the end 304 radially outwardly until it engages an outer bearing surface 310 of the hub 294, or shaft, of the pinion gear 338. The indentations 296 are formed and extend radially inwardly from the bearing surface 310. In one embodiment, the indentations include a ramped surface 298 tapering inwardly from and intersecting the bearing surface 310 and a stop surface 300 extending from and intersecting the ramped surface 298 toward and intersecting the bearing surface 310. The stop surface 300 is substantially orthogonal to a tangent of the bearing surface 310, and the intersection thereof, in one embodiment, although other angles may be suitable. The arm 302 biases the engagement end 304 into engagement with the indentation 296, and the surfaces 298 and 300 in particular, due to the preload. When the pinion gear 338 is rotated relative to the spur gear 339 in one direction, the arms 302, and the ends 304 thereof, will eventually engage the stop surface 300 as the arm is biased radially inwardly and thereafter drive the driven gear, or spur gear 339 with the drive gear 338 as the chain wheel 21 is rotated in one rotation direction, i.e., a pedaling direction. When the user backpedals, the chain 4 and chain pulley 21 are rotated in an opposite rotational direction, with the ends of the arms 302 successively and intermittently sliding outwardly along the ramp 298 as the arms 302 are biased out of engagement and thereafter along the bearing surface 310, such that the drive gear, or pinion gear 338, does not rotate the driven gear, or spur gear 339. In this way, the spring fingers 292, or curved arm 302, is biased into engagement with the indentation 296, and the stop surface 300, when the chain pulley 21 is rotated in the first rotational direction, and wherein the spring fingers 292, or curved arm 302, is biased out of engagement with the indentation 296 by the ramped surface 298 and bearing surface 310 when the chain pulley 21 is rotated in the second rotational direction such that the generator 50 is deactivated, or not rotated.

(62) Referring to FIGS. 14, 15, 31, 39 and 41, a belt tensioner 320 may be coupled to the cage assembly and is operable to adjust a tension of the belt 43. The belt tensioner 320 includes a first fastener 51, or first generator screw, which is threadably engaged with a first boss 24a formed on the outer cage 24, and a second fastener 52, or second generator screw, which is threadably

engaged with a second boss **24b** formed on the outer cage **24**. The first fastener **51** passes through an opening in the generator base **46**, and the second fastener **52** passes through a washer **53** overlapping an edge of the generator base **46**. The belt tensioner **320** further includes a belt tension adjustment member **54** that acts between the outer cage **24** and the generator **50**, and the base **46** in particular. In one embodiment, the adjustment member **54** may be configured as a set screw, which is threadably received in a boss **24b** formed in the outer cage **24**. An end **322** of the adjustment member **54** abuts the generator base **46**, for example an edge thereof. In operation, and in order to adjust the tension of the belt **43**, the fasteners **51**, **52** are left slightly loose. The adjustment member **54** may then be actuated, for example by threadably engaging the upper boss **24b**, such that the adjustment member **54** pushes on the edge of generator base **46** against a biasing force or tension of the belt **43**. The base **46** may rotate slightly around the fastener **51**, creating more tension in the belt **43**. When the desired belt tension has been reached, the fasteners **51**, **52** may be tightened, thus fixing the belt tension at the desired level. The adjustment member **54** may be provided with an anti-loosening device, for example a nylon patch to prevent loosening of the adjustment member **54** over time.

(63) Referring to FIGS. **15** and **19**, a plurality of generator wires **49**, shown as three, extend from the stator **48** of the generator. The generator wires **49** are electrically connected to a generator printed circuit board (generator PCB) **55**, which is mounted to the housing **200**. In one embodiment, the generator PCB **55** is secured to the outer cage **24** with a screw **56**, although the PCB **55** may be secured with adhesives, a snap fit or other suitable fasteners. The PCB **55** may alternatively be secured to the cover **23**.

(64) Referring to FIGS. **15**, **16**, and **19**, a main printed circuit board (main PCB) **57** is fixed relative to housing **200**, for example the cover **23**, by a first mounting screw **80** and a second motor holder mounting screw **78**. Referring to FIG. **16**, the main PCB **57** includes various electronic components, including but not limited to a power/energy storage device **57d**, configured in one embodiment as supercapacitors, a motor driver **57f**, an encoder **57e**, a switch **57b**, an LED **57c**, a plurality of pogo pins **57a**, and a microcontroller **402**. Referring to FIG. **15**, the pogo pins **57a** of the main PCB **57** are in electrical contact with the generator PCB **55**. The energy storage device **57d**, e.g., supercapacitors, also is coupled to and moveable with the cage assembly **9**, and in particular is secured to the main PCB **57**. The generator **50** generates energy which is transmitted to and stored in the energy storage device **57d**. In one embodiment, the energy storage device **57d** includes at least one capacitor, shown as two supercapacitors. In other embodiments, the energy storage device may include one or more batteries, for example rechargeable batteries.

(65) Referring to FIGS. **4**, **8** and **15**, a function button **19** is coupled to the housing **200**, and cover **23** in particular, by a button retainer **59**. An elastomeric seal **58** is disposed between the button retainer **59** and a surface of the cover **23**, or housing **200**. When the function button **19** is pressed by the user, the button **19** actuates the switch **57b** of the main PCB **57**. Among other functions, the function button **19** may be used to wirelessly pair the derailleur **180** with other system components, such as shifters that may be located remotely on the handle bars, for example. A LED lens **20** may be a clear lens that is fixed in a hole in the cover **23** or cage **24**, i.e. housing **200**. The LED lens **20** is positioned such that LED **57c** of the main PCB **57** shines through the lens **20** and is visible to the user. The LED **57c** may be used to indicate a state of the system. For example and without limitation, the LED **57c** is used when wirelessly pairing derailleur **180** with other system components, such as the shifters. The LED **57c** also may also be used to indicate other system states, such as battery life, for example. It should be understood that the generator may be positioned in other locations on the cage.

(66) As shown in FIGS. **38-41**, an electrical connector or conductor **502**, **502'**, configured in one embodiment as a wire or cable, electrically connects the electrical output of the generator **50** to an energy storage device **504**, **508**. The energy storage device **504** may be a high capacity capacitor **516**, one or more batteries **518** and/or a combination of a capacitor and/or batteries. The high

capacity capacitor **516** may be used for fast charging and discharging, while the battery **518**, or batteries, may be used for long term energy storage so the derailleur is ready to shift without the need to first charge the capacitors. The energy storage device **504** may include or house an electronic board **520** that controls the charging and discharging of either and/or both the capacitor **516** and/or the battery **518**. As shown in FIGS. **38-42**, the energy storage device **504**, **508** may be mounted or directly coupled to a component that is spaced apart from the cage assembly **9**, meaning the cage assembly **9** may move relative to the energy storage device **504**, **508**. The flexible electrical conductor **502** connects the generator system, or cage assembly **9**, to the energy storage device **504**, or mount **506** therefor. In one embodiment, the flexible electrical conductor is threaded through an opening **510** in the cage assembly, for example formed in an outer cage **24**, and thereafter into the interior of the housing where the flexible electrical conductor may be electrically coupled to the generator **50**, via the board **57**, as shown in FIG. **41**. As the cage assembly **9** moves relative to the energy storage device **504**, **508**, or mount **506** therefor, the flexible electrical conductor **502** flexes or deforms to accommodate relative movement therebetween. In this way the energy storage device **504**, **508** can be located on any other part or component of the derailleur spaced apart, or remote, from the cage assembly **9**, and which does not move along the same path as the cage assembly **9**. In this way, the terms “spaced apart” and “remote” refer to two components not being directly connected such that one component may, but does not necessarily, move, whether by way of translation or rotation, along the same path as the other component. For example, as shown in FIGS. **38-42**, the energy storage device **504**, **508** may be mounted on, or directly coupled to, the base **5**. Alternatively, as shown in FIG. **42**, the energy storage device may be mounted on, or directly coupled to, the linkage **186**, including either of the links **6**, **7**, or the moveable member **8**. In yet another embodiment, shown in FIG. **42**, the energy storage device **504**, **508** may not be mounted on any component of the derailleur, but rather is mounted to the frame **2**, for example the chain stay or seat stay or a mounting bracket attached to the frame with fastener **10**. Or, in other embodiments, the energy storage device may be configured to be successively mounted to any of the base **5**, linkage **186**, moveable member **8** or frame **2**, or a plurality of energy storage devices may simultaneously be mounted one or more of the base, linkage, moveable member and/or frame, with each of the plurality of energy storage devices being connected, via one or more conductors, to the generator system and/or motor, for example by way of the board **57**. In this way, and in these embodiments, the energy storage device **504**, **508** may be spaced apart from, or mounted remotely, to the cage assembly **9**.

(67) As shown in FIG. **39**, the flexible electrical conductor may be connected to the intermediate mount **506**, which may be coupled to any of the base **5**, coupling assembly **500**, including the linkage **186** or moveable member **8**, or the frame **2**. This mount **506** may be capable of removably and releasably receiving the energy storage device **508**, such as a battery or capacitor, and the mount **506** may or may not charge the energy storage device **506**. The mount may include one or more connectors, such as tabs, clips and/or ribs, which engage the energy storage device, for example with a snap fit. The energy storage device **508** coupled to the mount may include a standard battery that is not capable of being recharged by the generator. In this way the user may choose to use a traditional high capacity battery or a generator rechargeable storage device, e.g., battery or capacitor. The board **520** may control the charging of the capacitor **516** and/or rechargeable batteries **518**, and/or discharging of the batteries (chargeable or rechargeable) and/or capacitor to power a motor **60**. The board **520** may communicate with the main circuit board **57** to control the charging and discharging of the energy storage devices. The flexible electrical conductor **502** may be routed exteriorly to the derailleur components, as shown in FIG. **39**, or the flexible electrical conductor **502'** may be routed interiorly through at least a portion of the derailleur components, for example through interior portions of the coupling assembly **500**, including the linkage **186** and/or moveable member **8**. The mount **506** may include or incorporate an energy storage device **506**, such as a capacitor or rechargeable battery, as well as be configured

to connect to and engage an auxiliary energy storage device **508**, which may or may not be rechargeable.

(68) Motor and Shifting Drive System:

(69) FIGS. **17** and **18** depict a motor holder assembly **79**, including the motor **60**, which in one embodiment is coupled to and moveable with the cage assembly **9** in the opposite first and second directions. The motor **60** is operable to move the cage assembly **9** in the opposite first and second directions relative to the base member **5** and frame **2**, for example inboard and outboard lateral directions. A shifting drive system **330** is coupled between the motor **60** and the linkage **186**, wherein the shifting drive system includes an input from the motor **60** and an output coupled to the linkage **186**.

(70) The motor **60**, for example a DC motor, includes a first drive pinion gear **65** fixed to an output shaft **334**, which defines the input for the shifting drive system. The motor **60** is fixed to the motor holder **61** by at least one fastener **62**, for example a pair of screws. One end of a first drive axle **63** is received in a hole in motor holder **61**. A first drive spur gear **66** is coaxially fixed to a second drive pinion gear **67**, having a plurality of teeth (e.g., twenty (20)), and together the first drive spur gear **66** and the second drive pinion gear **67** rotate around the first drive axle **63**. The first drive pinion gear **65** has a plurality of teeth, e.g., twelve (12), spaced circumferentially around a periphery of the pinion gear that are engaged with a plurality of teeth (e.g., forth-eight (48)) spaced circumferentially around a periphery of a first drive spur gear **66**, providing a gear ratio of 4:1. An encoder gear **71**, having a plurality of teeth (e.g., thirty eight (38)), has a cylindrical recess in which an encoder magnet **72** is received. The encoder magnet **72** is fixed relative to encoder gear **71**. The encoder gear **71** has a long, cylindrical end portion that is rotatably received in a hole in motor holder **61**. The encoder gear **71** is engaged with the first drive spur gear **66**, and is axially retained by a retaining clip **73**. Referring to FIG. **19**, the motor holder assembly **79** is fixed to the cover **23** or outer cage **24**, i.e., housing **200**, with one or more fasteners, for example a first motor holder mounting screw **77** and a second motor holder mounting screw **78**. The encoder magnet **72** is positioned adjacent to encoder **57e** of the main PCB **57**. Electrical wires electrically connect the motor **60** to the main PCB **57**.

(71) Referring to FIG. **19**, the shifting drive system **330** includes a gear support plate **74** positioned relative to housing **200** by two positioning pins **76** that protrude from the cover **23**, and are retained by a screw **75**. The gear support plate **74** supports an end of the previously described first drive axle **63**. A second drive axle **64** has a first end that is received in an opening in the cover **23**, and a second end that is supported by the gear support plate **74**. A second drive spur gear **68** is coaxially fixed to a third drive pinion gear, with these two gears rotating together as a unit around the second drive axle **64**. The second drive spur gear **68** has a plurality of teeth (e.g., **48**) that are engaged with a plurality of teeth (e.g., 20) on the second drive pinion gear **67**, providing a gear ratio of 2.4:1. A third spur gear **70**, having a plurality of teeth (e.g., **42**), is engaged with the third drive pinion gear, having a plurality of teeth (e.g., 18), providing a gear ratio of 2.33:1. The shifting drive system has an overall gear ratio of $4 \times 2.4 \times 2.33 = 22.4:1$ in one embodiment, although other gear ratios may be suitable.

(72) Referring to FIGS. **19**, **20** and **24**, the shifting drive system **330** has an output with a second rotational speed that is less than the first rotational speed of an input, or the motor **60** and the shaft **334**, as determined by the overall gear ratio. In one embodiment, the output is configured as a lead screw **81**, which is coaxially fixed to the third spur gear **70**. The lead screw **81** is rotatable in first and second opposite rotation directions, and is threadably engaged with a crank arm **92**. The crank arm **92** is moveable in opposite first and second axial directions in response to the rotation of the lead screw **81** in the opposite first and second rotation directions. As further explained below, the crank arm **92** is coupled to the linkage **186**. In one embodiment, the lead screw **81** is partially threaded along its length. A lead screw bearing **83** is received in a recess formed in the cover **23**, or housing **200**. An adjustable bearing **84** is threadably received in the moveable member cover **93**,

which is fixed to the moveable member **8** with fasteners, shown as three screws **94**. The lead screw **81** is radially supported by a lead screw bearing **83** near a first end and is radially supported by an adjustable bearing **84** at a second end. Axial thrust loads of the lead screw **81** are reacted in a first direction by a thrust element **82** that is received in a recess in the outer cage **24**, and are reacted in a second direction by a surface of the adjustable bearing **84**. The adjustable bearing **84** may be threaded into the moveable member cover **93** to substantially eliminate any axial “play” of the lead screw **81**. Preferably, the adjustable bearing **84** includes a nylon locking element on its threaded portion to prevent it from moving after it has been set in the desired position.

(73) Referring to FIGS. **7** and **20-22**, a cage shaft **85** is fixed to the cover **23** and housing **200** with fasteners, for example six screws **86**. Alternatively, the cage shaft **85** may be fixed to the cover **23** by an over-molding process or by other means. Referring to FIG. **22**, the cage shaft **85** has two keying features **85a** that protrude radially inward from an inner diameter of cage shaft **85**. The cage shaft **85** is rotatably received in a hole in the moveable member **8** coaxial with the axis B and is axially retained to the moveable member **8** by a retaining ring **87**. A thrust washer **88** is disposed between the retaining ring **87** and a surface of the moveable member **8**. Alternatively, the thrust washer **88** may be replaced by shims of variable thickness that can be used to substantially eliminate any axial play between the cage shaft **85** and the moveable member **8**.

(74) Referring to FIGS. **20**, **21**, and **23**, a lead nut **91** has a threaded internal diameter and a first end **91b** having a spherical shape. Keyways **91a** are formed as grooves along the length of an outer diameter of lead nut **91**. Threads of the lead nut **91** are threadably engaged with threads of the lead screw **81**, and keying features **85a** of the cage shaft **85** are slidably received in keyways **91a** of the lead nut **91**. Rotation of the lead nut **91** relative to the cage shaft **85** is prevented by the key and keyway arrangement. Therefore, rotation of lead screw **81** causes translation of the lead nut **91** along axis B.

(75) Referring to FIGS. **20-21**, **25**, **26** and **27**, the crank arm **92** has a cylindrical recess **92a**, a first hole **92b** and a second hole **92c**. The crank arm **92** is rotatably connected to a fourth pin **18** at the second hole **92c**. The spherical end **91b** of lead nut **91** is engaged with the cylindrical recess **92a** of the crank arm **92**. The crank arm **92** includes a slot **92d**, which cuts through cylindrical recess **92a** and provides clearance for the elongated cylindrical portion of the lead nut **91** and lead screw **81**. Thus, contact between the lead nut **91** and crank arm **92** is maintained between the spherical portion **91b** of the lead nut **91** and the cylindrical recess **92a** of the crank arm **92**, which is preferably the only contact between those components in one embodiment.

(76) Referring to FIGS. **26-29**, a fourth pin **18** passes through a hole **99d** of a drive element **99**. In one embodiment, the drive element **99** may be fixed to the fourth pin **18** by a set screw **100** that is threadably engaged with a threaded hole **99e** of the drive element **99** and engages a flat surface **18a** formed on the fourth pin **18**. A drive pin **97** is received in the hole **92b** of the crank arm **92** and is fixed to the crank arm **92** by a set screw **104**. The drive pin **97** extends into a recess **99a** of the drive element **99**. A saver spring sleeve **96** is a cylindrical sleeve co-axially located relative to the fourth pin **18**. A saver spring **95** is a torsion spring that is located co-axially with the saver spring sleeve **96**. A first end of the saver spring **95** engages the crank arm **92**, and a second end of the saver spring **95** engages the surface **99c** of the drive element **99**. The saver spring **95** biases a surface **99b** of the drive element **99** against the drive pin **97** in the clockwise direction around an axis (C) shown in FIG. **20**.

(77) Referring to FIGS. **27-29**, the outer link **6** is fixed to the fourth pin **18** with a fastener, for example a set screw **101**, which engages a flat **18b** formed on the fourth pin **18**.

(78) Referring to FIG. **29**, a fastener **102**, for example a screw, is threadably engaged with an end of the fourth pin **18**. Prior to engaging the drive element **99** with the drive pin **97** disposed in the recess **99a** of drive element **99**, the saver spring **95** is in a “free,” or relaxed state. A tool, such as a screwdriver bit, may be engaged with the fastener **102**, e.g., by inserting the tool into a screw, and thereafter actuated to rotate the fourth pin **18** counterclockwise as shown in FIG. around the axis

(C). Since the fourth pin **18** is fixed to the drive element **99**, the drive element (**99**) is rotated with the pin **18**. The surface **99c** of the drive element **99** biases or pushes an end of the saver spring **95**, so as to load or wind up the saver spring **95** from a “free”, or relaxed, state to a preloaded state. The drive pin **97** may thereafter be slid into engagement with the recess **99a** of the drive element **99**. The tool may be removed, with the biasing force of the saver spring **95** biasing the surface **99b** of the drive element **99** clockwise around the axis (C) against the drive pin **97**.

(79) Referring to FIG. **20**, a cage spring **90** is formed as a torsion spring, a first end of which is disposed in a recess formed in the moveable member **8** and is engaged therewith. A second end of the spring **90** is engaged with the cover **23**, and is disposed in a recess in the cover **23**. The cage spring **90** is arranged to rotationally bias the cage assembly **9** in the clockwise direction around axis (B) as shown in FIG. **1** to take up any slack in the chain **4**.

(80) Again referring to FIG. **20**, a clutch spring **89** is shown as having a plurality of coils, for example in one embodiment with seven coils. The inner diameter of the a first plurality of coils, e.g., approximately the first five coils, is wrapped around a cylindrical surface of the moveable member **8**. The inner diameter of a second plurality of coils, or a single auxiliary coil, is wrapped around a cylindrical surface of the cage shaft **85**, with the first plurality of coils being greater than the second plurality (or auxiliary) coils. Because a greater number of coils is wrapped around the moveable member **8**, any relative rotation between the cage shaft **85** and the moveable member **8** may cause slippage between the clutch spring **89** and the cage shaft **85**, while the moveable member **8** and the clutch spring **89** remain fixed to each other. The clutch spring **89** is wound such that when the cage shaft **85**, together with the cage assembly **9**, rotates in the counterclockwise direction around axis (B), the resulting drag of the cage shaft **85** slipping against the clutch spring **89** causes the coils of the clutch spring **89** to tighten against the cylindrical surface of cage shaft **85**, which acts to increase friction and dampen the rotation of cage assembly **9**. When the cage shaft **85**, along with the cage assembly **9**, rotates in the clockwise direction around axis (B), the resulting drag of the cage shaft **85** against the clutch spring **89** causes the coils of the clutch spring **89** to loosen against the cylindrical surface of cage shaft **85**, which acts to reduce friction and allow the cage assembly **9** to rotate more easily. The function of the clutch spring **89** is to resist undesired (counterclockwise around axis (B)) rotation of the cage assembly **9**, which may allow the chain **4** to become slack, for example when the bicycle is traveling over rough terrain.

(81) As shown in FIGS. **38-41**, the motor **60** and drive system **330** may be located in the moveable member **8**. An output from the drive system **330** rotates the linkage **186** relative to the base member **5** and thereby moves the moveable member **8** relative to the base member **5**. In other embodiments, the motor **60** and drive system **330** may be mounted on or in the linkage **186**, including one or both of the links **6,7**.

(82) Operation:

(83) Referring to FIGS. **1, 2**, and **4**, when the rider pedals the bicycle, the chain **4** drives the cassette **3** clockwise, referring to FIG. **1**, and causes the rear wheel to rotate. Because the chain **4** is also engaged with upper chain pulley **21** and lower chain pulley **22**, the upper chain pulley **21** is driven counterclockwise, referring to FIG. **1**. Referring to FIGS. **10-13** and **38-41**, rotation of upper chain pulley **21** or lower chain pulley **22** causes the pulley shaft **35** to rotate. The first generator spur gear **37**, which is non-rotationally fixed to the pulley shaft **35**, is therefore also rotated. Referring to FIGS. **13, 39** and **41**, the transmission **250** provides for rotational power to flow from the first generator spur gear **37** to the first generator pinion gear **38** to the second generator spur gear **39** to the second generator pinion gear **40** to the first generator pulley **41** through the belt **43** to the second generator pulley **42** to the rotor **47** of the generator **50**. Rotation of the rotor **47** relative to the stator **48** generates electrical power. Referring to FIGS. **15** and **16**, the generated electrical power flows through the wires **49** of the generator **50** to the generator PCB **55**, and thereafter through the pogo pins **57a** into the main PCB **57**, where it is stored in the energy storage device **57d, 504, 508** for example the supercapacitors or one or more batteries. The energy storage device

57d, 504, 508 may thereafter supply power to the motor **60** to effect various shifting actions. It should be understood that the storage device **57d, 504, 508** may supply power to other electrical devices and accessories, including those located remote to the derailleur, for example and without limitation a wheel speed sensor, a cassette speed sensor, a power meter, lights, the front derailleur, an adjustable seat post, and/or other types of bicycle accessories, or may supply power to the motor and to such auxiliary devices. In embodiments where the electrical devices and accessories are located remote to the storage device, power may be transmitted by various electrical connectors or conductors, e.g., wires or cables. In other embodiments, the energy storage device **504, 508** also may be located remote to the derailleur, for example with the energy storage device mounted on another part of the frame **2** or bicycle component, and joined to the generator **50** with electrical connectors or conductors **502**, e.g., wires.

(84) It should be noted that the interface of the pulleys **41, 42** and the belt **43** results in less noise than an equivalent spur and pinion gear interface. Typically, gears that rotate relatively slowly do not create as much noise that may be objectionable to the rider, whereas gears that rotate at a high revolution per minute (RPM) may create a high-pitched noise that may be objectionable to the rider. Therefore, in one embodiment, the transmission may use gear interfaces when the angular velocities are relatively low, (i.e. close to the angular velocity of the upper pulley **21**), and a pulley belt interface may be used when the angular velocity of the pulleys are relatively high (i.e. close to the angular velocity of the rotor **47**).

(85) In order to request a gear shift, the rider operates a switch of a shifter on the handlebar. A wireless signal may be emitted from the shifter. The wireless signal is received by the antenna and radio on the main PCB **57**. The signal is processed by the main PCB **57**, and a controller provides for power to be transmitted from the power/energy storage device **57d, 504, 508**, e.g., the supercapacitors or battery, to the motor **60**. Referring to FIGS. **17** and **19**, mechanical power from the motor **60**, or a shifting drive system input, is transmitted from the first drive pinion **65** to the first drive spur **66** to the second drive pinion **67** to the second drive spur **68** to the third drive pinion to the third drive spur **70** to the lead screw **81**, or output.

(86) In the case that the rider requests a shift to a larger sprocket or cog of the cassette **3**, power supply provides power to the motor **60**, which rotates in the direction that, through the shifting drive system, including the drive pinion gears and spur gears **65-70**, drives the third drive spur gear **70** and the lead screw **81** in a first rotational direction. The lead nut **91** is threadably engaged with the lead screw, but the lead nut **91** is unable to rotate relative to the cage shaft **85**. Accordingly, the lead nut **91** moves axially along the axis (B) in the upwards direction, with reference to the orientation of FIG. **20**. The spherical end **91b** of the lead nut **91** drives the crank arm **92** counterclockwise around the axis (C) of the fourth pin **18**. Referring to FIGS. **28** and **29**, the drive pin **97** engages the surface **99b** of the drive element **99**, thereby rotating the drive element **99** counterclockwise around the axis (C) of the fourth pin **18**. Since the drive element **99** is fixed to the fourth pin **18**, the fourth pin **18** also rotates counterclockwise. Since the fourth pin **18** also is fixed to the outer link **6**, the outer link **6** also rotates counterclockwise around the axis (C), driving the linkage **186** in the inboard direction of the bicycle, which drives the moveable member **8** and the cage assembly **9** in the inboard direction. Since the chain **4** is engaged with upper chain pulley **21** of the cage assembly **9**, the chain **4** also is driven in the inboard direction, causing the chain **4** to shift to the next larger sprocket or cog of the cassette **3**.

(87) In the case that the rider requests a shift to a smaller sprocket or cog of the cassette **3**, the power supply supplies power to the motor **60**, which rotates in the direction that, through the shifting drive system, including the drive pinion gears and spur gears **65-70**, drives the third drive spur gear **70** and the lead screw **81** in a second rotational direction, driving lead nut **91** to move axially along axis (B) in the downwards direction. The spherical end **91b** of the lead nut **91** drives the crank arm **92** clockwise in FIG. **21** around the axis (C) of the fourth pin **18**. The crank arm **92** drives the saver spring **95** clockwise around axis (C) of the fourth pin **18**. The saver spring **95**

drives the drive element **99** clockwise around axis (C) of the fourth pin **18**. Since the drive element **99** is fixed to the fourth pin **18**, the fourth pin **18** also rotates clockwise. Since the fourth pin **18** is fixed to the outer link **6**, the outer link **6** also rotates clockwise around axis (C), driving the parallelogram linkage **186** in the outboard direction of the bicycle, which drives the moveable member **8** and the cage assembly **9** in the outboard direction. Since the chain **4** is engaged with upper chain pulley **21** of the cage assembly **9**, the chain **4** is also driven in the outboard direction, causing the chain **4** to shift into the next smaller sprocket or cog of the cassette **3**.

(88) Referring to FIGS. **16**, **17**, and **19**, the encoder **57e** on the main PCB **57**, together with encoder magnet (**72**), are used to provide positional feedback of the moveable member **8**, the cage assembly **9** and the chain pulley **21**. The encoder gear **71** is engaged with first drive spur gear **66**. As first drive spur gear **66** rotates, the gear **66** causes the encoder gear **71**, along with encoder magnet **72** to rotate about the axis of encoder gear **71**. Thus, the encoder magnet **72** rotates relative to the encoder **57e**. The encoder **57e** outputs a signal that indicates the angular position of encoder magnet **72** relative to the encoder **57e**. During a shifting operation, the encoder magnet **72** makes multiple complete revolutions. A controller on the main PCB **57** counts the number of times that encoder magnet **72** makes a complete revolution. Knowing the number of times that encoder magnet **72** has rotated, along with its current angular position relative to encoder **57e**, the controller can calculate the current position of the moveable member **8** relative to the bicycle frame **2**, and by extension the cassette **3**. A pre-determined shift table has multiple values, each of which corresponds to a shift position of derailleur **180**. As the encoder magnet **72** rotates during a shifting operation, a value corresponding to the current position of the moveable member **8** is compared to a target value in the table, with the controller capable or operable to determine when the moveable member **8** has reached its target position (i.e. the gear selected by the rider).

(89) In order for the system to establish a known relationship between the values of angular position output by the encoder **57e** and the physical location of the moveable member **8**, a homing routine may be performed. In this routine, the derailleur **180** positions the crank arm **92** at a reference position, and sets the corresponding angular position indicated by encoder **57e** as a reference value. Referring to FIG. **20**, the lead nut **91** is driven downwards, driving the crank arm **92** clockwise around axis (C) until contact between the crank arm **92** and a moveable member cover **93** prevents further clockwise rotation of the crank arm **92**. This is the reference position, which represents the farthest outboard position to which the derailleur can move. The corresponding positional value output by the encoder **57e** is established as a reference value. In one embodiment, this homing routine is performed at the factory when the derailleur **180** is assembled, but the routine may also be performed after derailleur **180** is installed on a bicycle. The homing routine may be performed even when the moveable member **8** is physically prevented from moving (i.e. when the rider isn't pedaling and the presence of the chain **4** prevents movement of the moveable member **8**). For example, in the event that the moveable member **8** is physically prevented from moving, the outer link **6**, fourth pin **18** and drive element **99** will also be prevented from moving. However, the crank arm **92** can still rotate clockwise to the position shown in FIG. **20** by winding up the saver spring **95**. Thus, the crank arm **92** may be driven to its reference position and the homing operation can be performed, even if movement of the moveable member **8** is blocked.

(90) The saver spring **95**, along with its associated parts, also acts as a safety clutch during impacts. In the event that the moveable member **8** or outer link **6** is acted upon by an external force that pushes the moveable member **8** in the inboard direction, the drive element **99** will be driven in the counterclockwise direction around axis (C). If the external force exceeds the preload of the saver spring **95**, the drive element **99** will wind up the saver spring **95**, preventing excessive force from being transmitted to the lead nut **91**. When the external force is removed, the saver spring **95** will unwind to its previous position.

(91) Control System and Generator Input/Rectification:

(92) Referring to FIG. 34, a schematic shows one embodiment of a control system for the generator 50, motor 60 and shifting drive system 330. As noted above, the chain pulley 21 or chain pulley 22 rotates the generator 50 which creates an alternating current (AC) voltage across each set of coils within the generator 50. The coils are connected to a rectification circuit 400 formed by diodes D1, D2, D3, D4, D5 and D6, which rectify the voltage to direct current (DC). The generator 50 may be configured as a permanent magnet brushless type generator having a plurality of (e.g., three) sets of coils internally connected preferably in a star configuration, although a delta configuration may also be suitable. The generator 50 may also be constructed from any number of coils so long as the coils are connected to the rectification circuit 400 that creates a DC voltage. The generator 50 may also be of a brushed type which creates DC voltage mechanically. Capacitors C3 and C4 are incorporated into the circuit to create a better impedance match to the energy storage device 57d, e.g., storage capacitors C1 and C2.

(93) Phase Sensing:

(94) Diodes D7, D8 and resistors R9, R10 are used to sense the speed and direction of the generator 50 which may be used to determine the speed and rotational direction of the chain pulley 21 and therefore the speed and rotational direction of the chain 4. A memory contains information about the number of teeth of each sprocket or cog of the cassette 3. As explained above, the encoder 57e provide a signal such that the controller, e.g., microcontroller 402, knows what gear is currently indexed. The microcontroller 402 is programmed with information about the number of teeth on chainring(s). If the bicycle has more than one chainring installed, the front chainring shifting system communicates the currently selected chainring to the rear derailleur 180 when the front chain ring is changed. With all this information, the controller 402 on the rear derailleur 180 knows how fast the bicycle is travelling and how fast the rider is turning the cranks. The bicycle speed and cadence information may be used to perform an automatic shifting algorithm which keeps the gear ratio at a preferable ratio for the rider. The diode anode of each diode-resistor pair is connected to a signal input on the microcontroller 402 as well as to the digital voltage supply Vdd through a pull up resistor. When the individual phase voltages of the diodes D7 or D8 drop below the VDD, the diode will conduct to ground and register a low signal to the microcontroller 402. When the phase voltage exceeds VDD the diode will not conduct and the signal to the microcontroller 402 will be clamped to VDD through R9 or R10. The sequence of pulses from each phase are different if the generator 50 is spinning clockwise or counter clockwise, as shown in FIGS. 35A and B, and any change in frequency is based on how fast the generator 50 is turning and therefore how fast the rider is pedaling. These sequences can be recognized by timing order by the microcontroller 402. The microcontroller 402 may modify the charging behavior if the pulse train corresponds to forward or reverse pedaling, for instance it may be desirable to disable charging, or deactivate the generator, while backpedaling, for example with the clutch, to reduce slack in the chain 4, which may cause the chain to disengage from the sprockets. It may be desirable to only allow charging above a certain rider cadence, and/or it may be desirable to always have charging enabled while not pedaling to reduce chain slap, etc. One way to determine rotational direction is to look at the pulses read by the microcontroller 402 for two adjacent generator coils. The phase of the L2 and L3 signals are separated by 120 degrees (of time) followed by 240 degrees and the sequence repeats when the generator 50 is spinning at a constant speed. The order of pulses indicates direction. For instance, an L2 signal followed by 120 degrees, then an L3 signal followed by 240 degrees and another L2 signal might indicate forward pedaling (FIG. 35A), where an L3 signal followed by 120 degrees then an L2 signal followed by 240 degrees followed by another L3 signal would indicate reverse pedaling (FIG. 35B). The actual generator speed is not likely to be perfectly constant in reality, but, because the generator 50 rotates many times for each turn of the cranks, analyzing the relative spacing of the L2, L3 sequence will be valid.

(95) Charge Control:

(96) Referring to FIG. 34, a P-channel mosfet Q2 connects and disconnects the rectified voltage

from the generator **50** to the storage device **57d** (e.g., capacitors **C1** and **C2**) for charging. Resistors **R6**, **R7**, **R8** and n-channel mosfets **Q3** and **Q4** allow the microcontroller **402** to control the **Q2** on/off state with a digital output while also setting **Q2** to a state by default if the voltage on **C1**, **C2** is too low to power the microcontroller **402**. Microcontroller **402** senses the state of charge of the capacitors (**C1**, **C2**) as a voltage through a voltage divider formed by **R11** and **R12**. The microcontroller **402** turns **Q2** off when maximum capacitor state of charge is reached and turns on **Q2** when capacitor state of charge falls below some threshold. The microcontroller **402** may switch **Q2** on and off rapidly with pulse width modulation or another variable duty cycle modulation scheme in order to regulate the effective rate of charge to reduce torque at the chain pulley. A proportional control of the **Q2** switching modulation may be implemented by the microcontroller such that a low state of charge of the storage capacitors (**C1**, **C2**) would cause **Q2** to be on at or nearly 100 percent charge rate but a state of charge close to maximum capacitor change may be at a very low charge rate. Modulation of the charge signal allows smooth transitions between not charging to charging, thereby preventing the user from feeling abrupt torque changes. As a redundancy to the microcontroller **402** turning off **Q2** at full charge a voltage sensing integrated circuit (IC) **408** may also disable **Q2** if the capacitor state of charge exceeds some threshold voltage. The threshold voltage that the voltage supervisor disables **Q2** is preferably higher than the voltage the microcontroller **402** would stop charging at by some small margin. If either the IC or the microcontroller **402** signals for charging to be disabled, **Q2** will turn off regardless of state of other control signal. Redundant charge control is useful for situations where the microcontroller **402** is unable to perform charge control function, for instance during a firmware update of the microcontroller. An additional redundant overvoltage protection of the capacitors is optionally provided by an integrated circuit (IC) **U4**, which may be of type LM317 shunt regulator. This IC senses capacitor voltage and turns on **Q1** shorting the capacitors to ground if a safe voltage threshold is exceeded, which will discharge capacitors until capacitor voltage is below safe threshold. Capacitors (**C1**, **C2**) are preferably of electrochemical double-layer (ELDC) type also known as super capacitors. The schematic shows two capacitors (**C1**, **C2**) in series but any number of series capacitors may be used. The number of capacitors in series should be matched to the preferred operating voltage of the motor **60** and open circuit voltage of the generator **50**. The capacitors **57d** may also be arranged in a multiple parallel-series arrangement to achieve different voltage and capacitance values. The capacitors (**C1**, **C2**) may be of other types such as ceramic, electrolytic, tantalum or a future capacitor technology. The capacitors (**C1**, **C2**) function may also be performed by a rechargeable battery such as a lithium ion battery or lithium polymer battery or some future battery technology. Resistors **R1** and **R2** function to balance the charge between the two capacitors over time so that one capacitor does not develop an excess of charge compared to the other capacitor.

(97) Wrapping Encoder:

(98) In one embodiment of the derailleur **180**, configured with the linkage **186** actuated by lead screw **81**, the encoder **57e** may be required to count multiple turns of sensing element (such as a diametrically magnetized magnet). This turn counting can be performed by the encoder **57e** or by the microcontroller **402** noting when the encoder wraps over its max value. The absolute position of the linkage **186** is determined by moving the derailleur to one end of possible travel and detecting a stall condition on the motor **60**. Ideally this motion limit detection is performed only once during manufacturing, therefore the encoder system must store the number of full rotational wraps between power cycles of the device. In one embodiment, the derailleur **180** has the characteristic of having a non-removable power supply **57d** (capacitors (**C1**, **C2**)), meaning the power source cannot be suddenly disconnected.

(99) Referring to FIG. **36**, when the bicycle is not being ridden, the capacitor voltage may slowly fall due to internal leakage, the balancing resistors and the small power requirements of the microcontroller **402**, and may drop to 0. When the voltage of the capacitor **57d** approaches a

threshold voltage (e.g., 3.5V), or a voltage slightly above the minimum voltage required to power the encoder **57e**, (e.g., 3.3V), the microcontroller **402** may save the current wrap count to the memory. As shown in FIG. **36**, when the voltage of the capacitors **57d** starts increasing again due to the bike being pedaled, the microcontroller will turn on when the voltage reaches a first threshold voltage (e.g., 1.8V), the encoder will turn on when the voltage reaches a second threshold voltage (3.3V), and at some voltage threshold (e.g., 3.7V), higher than the voltage at which the derailleur saved the last wrap count, but lower than a threshold voltage (e.g., 4.2V) at which the microcontroller **402** allows shifting to occur, the microcontroller **402** will read the current encoder position and add the last known wrap count to restore a knowledge of the absolute position of the derailleur. Above the shifting threshold voltage (e.g., 4.2V), shifting may occur.

(100) Referring to FIG. **34**, a voltage regulator **410** may be included to provide a voltage source lower than the voltage of the capacitor **57d** to supply the microcontroller **402**, encoder **57e**, motor controller and other ICs in the circuit. The voltage regulator **410** may be integral to the microcontroller **402**, or one of the other ICs, which in turn supplies a lower voltage to other components in the circuit. The voltage regulator **410** is preferably a switching buck type, but may also be a linear type or buck boost type. The voltage regulator **410** may be omitted if the capacitor voltage does not exceed the maximum ratings for the microcontroller or other ICs.

(101) A switch **SW1** is connected to a digital input of the microcontroller **402** to provide user input to system such as wireless pairing function. More or fewer switches may be used for various user input.

(102) Optional LEDs **D9**, **D10** and **D11** may be provided to indicate to the user about the derailleur **180** such as pairing state or state of capacitor charge. These may be omitted. The LEDs may be configured to illuminate in various colors.

(103) Motor driver **57f** is operatively connected to the microcontroller **402** by an appropriate control signal or set of control signals such as an analog control signal, a digital control signal such as I2C, SPI, UART, etc., or may be control by pulse width modulated digital signals.

(104) The microcontroller **402** may include a general processor, digital signal processor, an application specific integrated circuit (ASIC), field programmable gate array (FPGA), analog circuit, digital circuit, combinations thereof, or other now known or later developed processor. The processor may be a single device or combinations of devices, such as through shared or parallel processing.

(105) The memory may be a volatile memory or a non-volatile memory. The memory may include one or more of a read only memory (ROM), random access memory (RAM), a flash memory, an electronic erasable program read only memory (EEPROM), or other type of memory. The memory may be removable from the derailleur **180**, such as a secure digital (SD) memory card. In a particular non-limiting, exemplary embodiment, a computer-readable medium can include a solid-state memory such as a memory card or other package that houses one or more non-volatile read-only memories. Further, the computer-readable medium can be a random-access memory or other volatile re-writable memory. Additionally, the computer-readable medium can include a magneto-optical or optical medium, such as a disk or tapes or other storage device. Accordingly, the disclosure is considered to include any one or more of a computer-readable medium and other equivalents and successor media, in which data or instructions may be stored.

(106) The memory may be a non-transitory computer-readable medium and may be described as a single medium. However, the term “computer-readable medium” includes a single medium or multiple media, such as a centralized or distributed memory structure, and/or associated caches that are operable to store one or more sets of instructions and other data. The term “computer-readable medium” shall also include any medium that is capable of storing, encoding or carrying a set of instructions for execution by a processor or that cause a computer system to perform any one or more of the methods or operations disclosed herein.

(107) In an alternative embodiment, dedicated hardware implementations, such as application

specific integrated circuits, programmable logic arrays and other hardware devices, can be constructed to implement one or more of the methods described herein. Applications that may include the apparatus and systems of various embodiments can broadly include a variety of electronic and computer systems. One or more embodiments described herein may implement functions using two or more specific interconnected hardware modules or devices with related control and data signals that can be communicated between and through the modules, or as portions of an application-specific integrated circuit. Accordingly, the present system encompasses software, firmware, and hardware implementations.

(108) A communication interface provides for data and/or signal communication from and between the derailleur to another component of the bicycle, or an external device such as a mobile phone or other computing device. The communication interface communicates the data using any operable connection. An operable connection may be one in which signals, physical communications, and/or logical communications may be sent and/or received. An operable connection may include a physical interface, an electrical interface, and/or a data interface. The communication interface may be configured to communicate wirelessly, and as such include one or more antennae. The communication interface provides for wireless communications in any now known or later developed format. Although the present specification describes components and functions that may be implemented in particular embodiments with reference to particular standards and protocols, the invention is not limited to such standards and protocols. For example, standards for Internet and other packet switched network transmission (e.g., TCP/IP, UDP/IP, HTML, HTTP, HTTPS) represent examples of the state of the art. Such standards are periodically superseded by faster or more efficient equivalents having essentially the same functions. Bluetooth® and or ANT+™ standards may also, or alternatively, be used. Accordingly, replacement standards and protocols having the same or similar functions as those disclosed herein are considered equivalents thereof.

(109) The communication interface is configured to send and/or receive data such as control signals and/or commands to and/or from bicycle components such as the front gear changer and/or the derailleurs **180**. The component communication interface communicates the data using any operable connection. An operable connection may be one in which signals, physical communications, and/or logical communications may be sent and/or received. An operable connection may include a physical interface, an electrical interface, and/or a data interface. The communication interface provides for wireless communications in any now known or later developed format. Although the present specification describes components and functions that may be implemented in particular embodiments with reference to particular standards and protocols, the invention is not limited to such standards and protocols. For example, standards for Internet and other packet switched network transmission (e.g., TCP/IP, UDP/IP, HTML, HTTP, HTTPS) represent examples of the state of the art. Such standards are periodically superseded by faster or more efficient equivalents having essentially the same functions. Accordingly, replacement standards and protocols having the same or similar functions as those disclosed herein are considered equivalents thereof.

(110) In accordance with various embodiments of the present disclosure, methods described herein may be implemented with software programs executable by a computer system, such as the microcontroller and circuitry. Further, in an exemplary, non-limited embodiment, implementations can include distributed processing, component/object distributed processing, and parallel processing. Alternatively, virtual computer system processing can be constructed to implement one or more of the methods or functionality as described herein.

(111) A computer program (also known as a program, software, software application, script, or code) can be written in any form of programming language, including compiled or interpreted languages, and it can be deployed in any form, including as a standalone program or as a module, component, subroutine, or other unit suitable for use in a computing environment. A computer program does not necessarily correspond to a file in a file system. A program can be stored in a

portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, sub programs, or portions of code). A computer program can be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.

(112) The processes and logic flows described in this specification can be performed by one or more programmable processors executing one or more computer programs to perform functions by operating on input data and generating output. The processes and logic flows can also be performed by, and apparatus can also be implemented as, special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit).

(113) As used in this application, the term ‘circuitry’ or ‘circuit’ refers to all of the following: (a) hardware-only circuit implementations (such as implementations in only analog and/or digital circuitry) and (b) to combinations of circuits and software (and/or firmware), such as (as applicable): (i) to a combination of processor(s) or (ii) to portions of processor(s)/software (including digital signal processor(s)), software, and memory(ies) that work together to cause an apparatus, such as a mobile phone or server, to perform various functions) and (c) to circuits, such as a microprocessor(s) or a portion of a microprocessor(s), that require software or firmware for operation, even if the software or firmware is not physically present.

(114) This definition of ‘circuitry’ applies to all uses of this term in this application, including in any claims. As a further example, as used in this application, the term “circuitry” would also cover an implementation of merely a processor (or multiple processors) or portion of a processor and its (or their) accompanying software and/or firmware, as well as other electronic components. The term “circuitry” would also cover, for example and if applicable to the particular claim element, a baseband integrated circuit or applications processor integrated circuit for a mobile computing device or a similar integrated circuit in server, a cellular network device, or other network device.

(115) Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor receives instructions and data from a read only memory or a random-access memory or both. The essential elements of a computer are a processor for performing instructions and one or more memory devices for storing instructions and data. Generally, a computer also includes, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto optical disks, or optical disks. However, a computer need not have such devices. Moreover, a computer can be embedded in another device, e.g., a mobile telephone, a personal digital assistant (PDA), a mobile audio player, a Global Positioning System (GPS) receiver, or a power meter system to name just a few. Computer readable media suitable for storing computer program instructions and data include all forms of non-volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and CD ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

(116) The illustrations of the embodiments described herein are intended to provide a general understanding of the structure of the various embodiments. The illustrations are not intended to serve as a complete description of all of the elements and features of apparatus and systems that utilize the structures or methods described herein. Many other embodiments may be apparent to those of skill in the art upon reviewing the disclosure. Other embodiments may be utilized and derived from the disclosure, such that structural and logical substitutions and changes may be made without departing from the scope of the disclosure. Additionally, the illustrations are merely representational and may not be drawn to scale. Certain proportions within the illustrations may be

exaggerated, while other proportions may be minimized. Accordingly, the disclosure and the figures are to be regarded as illustrative rather than restrictive.

(117) While this specification contains many specifics, these should not be construed as limitations on the scope of the invention or of what may be claimed, but rather as descriptions of features specific to particular embodiments of the invention. Certain features that are described in this specification in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination.

(118) Similarly, while operations and/or acts are depicted in the drawings and described herein in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the embodiments described above should not be understood as requiring such separation in all embodiments, and it should be understood that any described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

(119) One or more embodiments of the disclosure may be referred to herein, individually and/or collectively, by the term “invention” merely for convenience and without intending to voluntarily limit the scope of this application to any particular invention or inventive concept. Moreover, although specific embodiments have been illustrated and described herein, it should be appreciated that any subsequent arrangement designed to achieve the same or similar purpose may be substituted for the specific embodiments shown. This disclosure is intended to cover any and all subsequent adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, are apparent to those of skill in the art upon reviewing the description.

(120) The Abstract of the Disclosure is provided to comply with 37 C.F.R. § 1.72(b) and is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, various features may be grouped together or described in a single embodiment for the purpose of streamlining the disclosure. This disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter may be directed to less than all of the features of any of the disclosed embodiments. Thus, the following claims are incorporated into the Detailed Description, with each claim standing on its own as defining separately claimed subject matter.

(121) It is intended that the foregoing detailed description be regarded as illustrative rather than limiting and that it is understood that the following claims including all equivalents are intended to define the scope of the invention. The claims should not be read as limited to the described order or elements unless stated to that effect. Therefore, all embodiments that come within the scope and spirit of the following claims and equivalents thereto are claimed as the invention.

(122) Although embodiments have been described for illustrative purposes, those skilled in the art will appreciate that various modifications, additions, and substitutions are possible, without departing from the scope and spirit of the disclosure. It is therefore intended that the foregoing description be regarded as illustrative rather than limiting, and that it be understood that all equivalents and/or combinations of embodiments and examples are intended to be included in this description.

(123) Although certain parts, components, features, and methods of operation and use have been

described herein in accordance with the teachings of the present disclosure, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all embodiments of the teachings of the disclosure that fairly fall within the scope of permissible equivalents.

Claims

1. A bicycle derailleur comprising: a base member mountable to a bicycle frame; a cage assembly moveably coupled to the base member with a coupling assembly, the cage assembly including an upper pulley and a lower pulley; an electrical generator system coupled to and moveable with the cage assembly, the electrical generator system including a generator drive system configured with the upper pulley as an input; a motor coupled to and moveable with the cage assembly, wherein the motor is operable to move the cage assembly in opposite first and second directions; and at least one energy storage device mounted on at least one of the base member and/or coupling assembly, wherein the energy storage device is electrically connected to the electrical generator system.
2. The bicycle derailleur of claim 1 wherein the coupling assembly comprises a linkage rotatably connected to the base member and a moveable member rotatably connected to the linkage, wherein the cage assembly is rotatably coupled to the moveable member, and wherein the energy storage device is mounted on at least one of the base member, the linkage and/or the moveable member.
3. The bicycle derailleur of claim 2 wherein the energy storage device is mounted on the base member.
4. The bicycle derailleur of claim 1 wherein the energy storage device is electrically connected to the electrical generator system with a flexible electrical conductor.
5. The bicycle derailleur of claim 4 wherein the flexible electrical conductor is routed interiorly through at least a portion of the coupling assembly.
6. The bicycle derailleur of claim 4 further comprising a mount mounted directly to the at least one of the base member and/or coupling assembly, wherein the mount is electrically connected to the electrical generator system with the flexible electrical conductor, and wherein the energy storage device is releasably coupled to the mount.
7. The bicycle derailleur of claim 1 wherein the energy storage device comprises a capacitor.
8. The bicycle derailleur of claim 1 wherein the energy storage device comprises a battery.
9. The bicycle derailleur of claim 8 wherein the battery is rechargeable by the electrical generator system.
10. The bicycle derailleur of claim 1 wherein the electrical generator system supplies power to the motor.
11. The bicycle derailleur of claim 10 wherein the electrical generator system comprises a generator, wherein the generator generates energy stored in the energy storage device, and wherein the motor is electrically coupled to, and powered by, the energy storage device.
12. A bicycle derailleur comprising: a base member mountable to a bicycle frame; a moveable member movably connected to the base member; a cage assembly rotatably coupled to the moveable member, the cage assembly moveable relative to the moveable member in a first direction and a second direction, the cage assembly including an upper pulley and a lower pulley; an electrical generator system coupled to and moveable in the first direction and the second direction with the cage assembly, the electrical generator system operated by rotation of the upper pulley; a motor coupled to and moveable with the cage assembly in the first and the second direction, wherein the motor is operable to move the cage assembly in the first and the second direction; an energy storage device spaced apart from the cage assembly, wherein the cage assembly is moveable relative to the energy storage device; and a flexible electrical conductor electrically connecting the energy storage device and the electrical generator system.
13. The bicycle derailleur of claim 12 further comprising a linkage rotatably connected to the base member and the moveable member rotatably connected to the linkage, wherein the energy storage

device is mounted on at least one of the base member, the linkage and/or the moveable member.

14. The bicycle derailleur of claim 13 wherein the energy storage device is mounted on the base member.

15. The bicycle derailleur of claim 12 wherein the flexible electrical conductor is routed interiorly through at least a portion of a coupling assembly.

16. The bicycle derailleur of claim 12 further comprising a mount releasably coupled to the energy storage device, wherein the mount is electrically connected to the flexible electrical conductor.

17. The bicycle derailleur of claim 12 wherein the energy storage device comprises a battery.

18. The bicycle derailleur of claim 17 wherein the battery is rechargeable by the electrical generator system.

19. The bicycle derailleur of claim 12 wherein the electrical generator system supplies power to the motor.

20. The bicycle derailleur of claim 19 wherein the electrical generator system comprises a generator, wherein the generator generates energy stored in the energy storage device, and wherein the motor is electrically coupled to, and powered by, the energy storage device.
